

Data Cleaning Rules

Cornell 1B | AI Companion Risk Radar | Break Through Tech AI Studio, Fall 2026

Purpose

This document records how the MHARD dataset was cleaned and narrowed before hand labeling. Every rule below lists what was done and why, so the workflow can be reproduced and so the team can explain its decisions to the Challenge Advisor and in the final presentation.

Starting point

Item	Value
Source file	MHARD_dataset.csv (Sensify Lab)
Total reviews	200,972
Apps	73 mental health apps on Google Play
Date range	March 29, 2011 to July 11, 2023
Reviews after selection and cleaning	18,801 across 9 apps

Rules applied

Rule 1. Keep only AI and chatbot relevant apps

The project is about AI companion risks, so only apps where users can talk to an AI were kept. Nine apps were selected out of 73 and sorted into two tiers.

- **Tier 1, the app is primarily a chatbot:** wysa, woebot, youper
- **Tier 2, the app has an AI chat feature alongside other tools:** amaha, happify, healthily, vos, voidpet, hey

Apps were chosen by measuring how often each app's reviews mention bot, chatbot, AI, or robot, then reading samples of those reviews to confirm the mentions referred to an AI the user talks to. Apps were excluded when their mentions turned out to be about moderation bots (talklife), human listeners described as robotic (sevincups, ifeel), or a cartoon mascot (reflectly, finch). Pending confirmation from the Challenge Advisor on whether to use Tier 1 only, both tiers, or both with a comparison.

Rule 2. Drop reviews with no text

Twenty one reviews in the full dataset have an empty review field. They cannot be labeled or used for text classification.

Rule 3. Drop duplicate review text within the same app

Where the same app has the same review text more than once, the first copy is kept and the rest removed. The full dataset has 6,189 such duplicates. Identical text under two different apps is kept,

since each is a real review of that app.

Rule 4. Drop reviews of three words or fewer

Reviews like good app or love it are too short to express a risk theme. The cutoff at four words is a judgment call and can be revisited.

Rule 5. Use the raw review column, not review_cleaned

This is the most important rule. The review_cleaned column has stopwords removed, including not, no, me, and myself. Those words carry the meaning the risk themes depend on. I have no one to talk to becomes talk once cleaned, which erases the isolation signal. All labeling and modeling should use the raw review column.

Rule 6. Convert the date column to datetime

Dates are stored as text such as October 27, 2018. Converting them allows sorting, filtering by year, and the weekly or monthly aggregation the temporal dashboard stretch goal needs.

Rule 7. Keep all ratings

No rating is excluded. Risk themes appear across the full range, and many attachment and dependence reviews are five stars because the user is happy with the app.

What was deliberately not done

- **No stopword removal or negation stripping before labeling.** The project brief warns against removing words that carry meaning for dependence, attachment, isolation, or frustration. Stopword handling can be tested later at the TF-IDF stage.
- **No lowercasing or punctuation stripping in the stored text.** TF-IDF lowercases during modeling, so the saved text stays readable for the people doing the labeling.
- **No removal of emoji or non English reviews yet.** These should be counted first, then handled as a documented decision.
- **No filtering by rating, likes, or review length beyond the three word floor.**

Open questions for the team

- **Non English reviews.** MHARD includes reviews in other languages. TF-IDF handles mixed languages poorly and the keyword patterns are English only. Options are to keep them and state the limitation, drop them, or label English only. Count them before deciding.
- **The word count floor.** Four words is the current cutoff. A five word review such as it is my only friend is short but meaningful, so a lower floor or a character count may work better.
- **Near duplicates.** The current rule catches exact matches only. Reviews differing by one character remain in the data.
- **Developer responses.** The response column is unused. The team should confirm that labels describe what the user wrote, not the exchange with the developer.
- **Subset balance.** Amaha is 27 percent of the subset at 5,115 reviews even though it is a Tier 2 app, while wysa, the most chatbot focused app, is 1,790. This affects how the labeling sample should be

drawn and belongs in the limitations section of the final presentation.

Reproducing this

All rules are implemented in the data exploration notebook in the team repository. The notebook loads MHARD_dataset.csv, prints the counts behind each rule, and writes two files: ai_subset_clean.csv, the 18,801 cleaned reviews, and pilot_labeling_sample_200.csv, a shuffled sample of 200 reviews prepared for pilot labeling. Random operations use a fixed seed of 42 so anyone running the notebook gets the same sample.